

Question 1: What is Power BI and why is it used in businesses?

-> Microsoft Power BI is a business analytics and data visualization tool developed by Microsoft. It helps users connect to different data sources, transform data, create interactive dashboards, and generate reports.

Why businesses use Power BI:

- To analyze large amounts of data easily
- To create visual dashboards and reports
- To make better business decisions using insights
- To track business performance in real time
- To share reports across teams and departments
- To automate reporting processes

Question 2: Name and explain the three main components of Power BI.

-> 1. Power BI Desktop

It is a desktop application used to:

- Import and clean data
- Create data models
- Design reports and dashboards

2. Power BI Service

It is a cloud-based online service used to:

- Publish reports
- Share dashboards with others
- Collaborate within organizations
- Access reports from anywhere

3. Power BI Mobile

It is a mobile application used to:

- View reports and dashboards on smartphones/tablets
- Monitor business performance on the go
- Receive alerts and notifications

Question 3: Explain the Power BI workflow.

-> The Power BI workflow consists of the following steps:

1. Connect Data
 - Import data from Excel, SQL, CSV, Web, etc.
2. Transform Data
 - Clean and modify data using Power Query.
3. Create Data Model
 - Build relationships between tables.
4. Visualize Data
 - Create charts, graphs, dashboards, and reports.
5. Publish Reports
 - Upload reports to Power BI Service.
6. Share and Collaborate
 - Share dashboards with team members.
7. Monitor and Update
 - Refresh data regularly for updated insights.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

-> Four data cleaning tasks in Power Query are:

1. Removing duplicate rows
2. Replacing missing or null values
3. Changing data types (Text, Number, Date, etc.)
4. Splitting columns into multiple columns

Other tasks include filtering rows, merging tables, renaming columns, and trimming spaces.

Question 5 : Write step by step instructions to load the above dataset.

Dataset Used

Dataset Name: Real_Estate_Data_100_Rows.csv

The dataset contains 100 rows and 10 columns:

- Property_ID
- Property_Name
- City
- Location_Details
- Price
- Area_sqft
- Bedrooms
- Bathrooms
- Owner_Name
- Contact

Steps to Load Dataset in Power BI:

1. Open Power BI Desktop.
2. Click on Home -> Get Data -> Text/CSV.
3. Browse and select Real_Estate_Data_100_Rows.csv.
4. Click Open.
5. Preview the dataset in the Navigator window.
6. Click Transform Data to clean the dataset.
7. Verify columns such as Price, Area_sqft, Bedrooms, and Bathrooms.
8. Click Close & Apply to load the cleaned dataset into Power BI.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

-> Data View

Data View is used to:

- View table data in rows and columns
- Inspect imported data
- Create calculated columns
- Analyze records directly

Purpose:

It helps users understand and verify the dataset.

2. Report View

Report View is the main workspace used to:

- Create visualizations
- Design dashboards
- Add charts, graphs, maps, and tables
- Apply filters and slicers

Purpose:

It is used for data visualization and report creation.

3. Model View

Model View displays relationships between tables.

Purpose:

- Create relationships between datasets
- Manage data models
- Improve data analysis
- Build star or snowflake schemas

Question 7 : Discuss the different data sources that Power BI supports.

-> Power BI supports many types of data sources, including:

File Sources

- Excel
- CSV
- XML
- JSON
- PDF

Database Sources

- SQL Server
- MySQL
- Oracle Database
- PostgreSQL
- MongoDB

Cloud Services

- Google Analytics
- Azure
- Salesforce
- SharePoint

Online Services

- Web APIs
- Websites
- OData feeds

Big Data Sources

- Hadoop
- Spark
- Azure Data Lake

Power BI can combine multiple data sources into one report.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

-> The dataset contains an Owner_Name column with full names such as:

- Rahul Mehta
- Anita Sharma
- Ramesh Kumar
- Neha Verma

Steps to Split Owner_Name Column:

1. Open Power Query Editor in Power BI.
2. Select the Owner_Name column.
3. Go to the Transform tab.
4. Click Split Column -> By Delimiter.
5. Choose Space as the delimiter.
6. Select Split at left-most delimiter.
7. Click OK.

Two new columns will be created:

- First Name
- Last Name